

The average investor, who holds the market portfolio by definition, does not rebalance.

4. Investors have *homogeneous* expectations.

This assumption ensures that all investors hold the same MVE portfolio in the CAPM world and that, in equilibrium, the MVE portfolio becomes the market portfolio. In the real world, though, people obviously do not all share the same beliefs; they have *heterogeneous* expectations. By itself, the homogeneous expectations assumption is not important: a version of the CAPM holds where the expected returns are simply averages across the beliefs of all investors.¹² But, in combination with the next assumption, heterogeneous expectations can produce significant deviations from the CAPM.

5. No taxes or transactions costs.

Taxes matter. Chapter 12 shows that taxes affect expected returns and can be regarded as a systematic factor. Transactions costs, meanwhile, also vary across securities. We should expect that for very illiquid markets with high transactions costs, there may be more deviations from the CAPM. This is indeed the case, and chapter 13 discusses various liquidity premiums in more detail.

There is another effect of transaction costs when trading frictions are combined with heterogeneous investors. If investors cannot short, then investor beliefs matter. Optimists may prevail in pricing because the pessimists' beliefs are not impounded into stock prices. Pessimists would like to short but cannot, and so stock prices reflect only the belief of optimists. Thus, investor beliefs become a systematic factor. While there are behavioral versions of this story, the original setting of Miller (1977), where this concept was developed, was a rational setting. Related to this assumption is the next one, since when individuals move prices, markets are likely to be illiquid and there are many trading frictions:

6. Individual investors are price takers.

The informed investor is trading and moving prices because he has some knowledge that others do not have. But when these trades are large, they move prices, which leads us to. . . .

7. Information is costless and available to all investors.

Processing and collecting information is not costless, and certain information is not available to all investors. Information itself can be considered a factor in some economic settings, as in Veldkamp (2011). The CAPM applies in a stylized, efficient market; we should think that additional risk premiums can be collected in more inefficient securities markets, especially where information is very costly and not available to many investors. As we explore in the next few chapters, this is indeed the case; several deviations from the

¹² See Williams (1977).